APPLICATION NOTE

RFSW8000 5MHz to 6500MHz 10W SPDT Switch

Introduction

This application note explains the operation of the RFSW8000 5 MHz to 6500 MHz Single-Pole Double-Throw Switch. The RFSW8000 is a high-power switch designed for high performance IEEE802.11 a/b/g/n/ac, 802.15.4, 802.15.1 and sub 1 GHz wireless systems where extremely high linearity, high isolation, low insertion loss, and small package size are required. It is nominally battery operated from +2.5V to +5.0V while drawing 5 µA. It provides 0.40 to 0.85 dB Insertion Loss with 25 - 29 dB of isolation. It supports excellent linearity with an IIP3 of +59dBm. This SPDT switch is a 50-ohm part and is housed in an 8-pin 2.0 x 2.0 mm DFN package. For more detailed information, please refer to the RFSW8000 datasheet.

Product Details

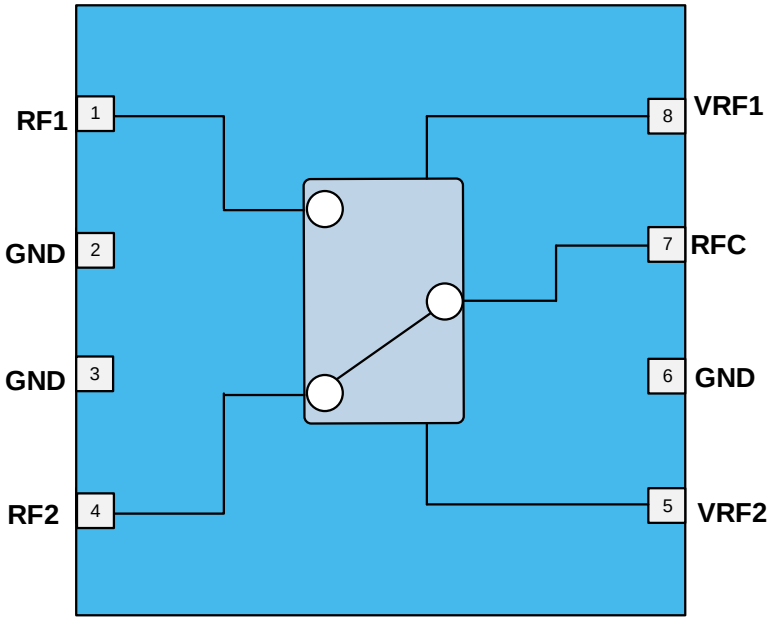


Figure 1. Functional Block Diagram & Pin-out Detail

Table 1. RFSW8000 Pin Description

PIN NUMBER	LABEL	DESCRIPTION
1	RF1	RF input. Internally matched to 50 Ω and DC coupled
2, 3, 6	GND	Ground connection
4	RF2	RF input. Internally matched to 50 Ω and DC coupled
5	VRF2	Control Voltage
7	RFC	RF common port. Internally matched to 50 Ω and DC coupled
8	VRF1	Control Voltage
GND Paddle	Package Base	RF/DC ground connection. Use recommended via pattern to minimize inductance and thermal resistance. See PCB Mounting Pattern for suggested footprint

Evaluation Board Information

The Qorvo RFSW8000 Evaluation Board (EVB) is designed to provide performance representative of that obtainable in an actual application. The EVB is designed to operate with 50 Ω load impedances at all RF ports, which are provided with SMA connector interfaces. Only a series capacitor for AC coupling is applied between the module RF pins and the EVB SMA connectors.

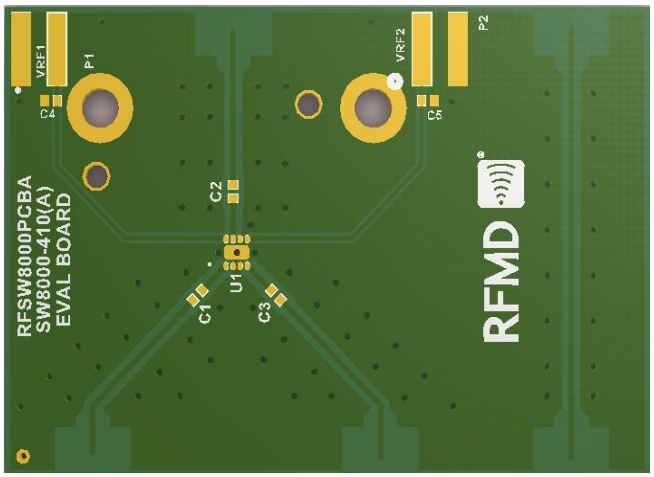


Figure 2a. RFSW8000 Evaluation Board Photo

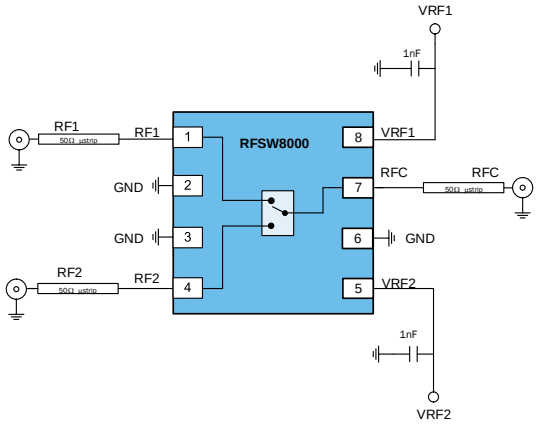


Figure 2b. RFSW8000-EVB Schematic

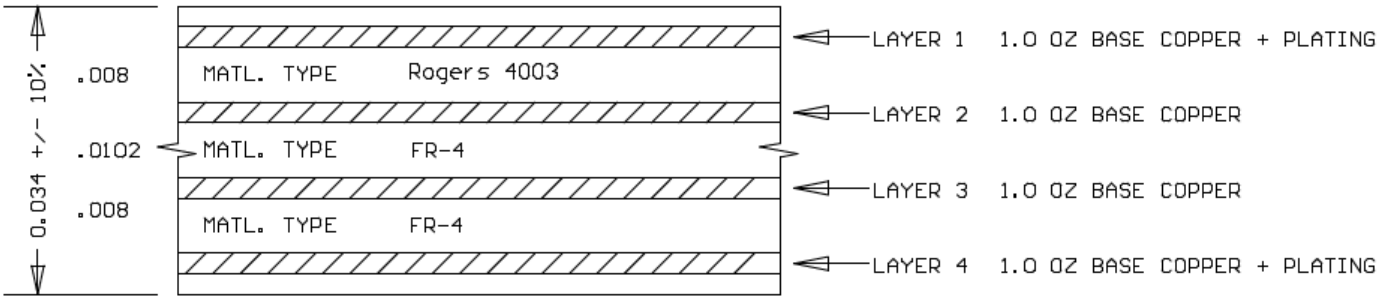


Figure 2c. RFSW8000-PCB Stack-up

Recommended Biasing Sequence

Table 2 below provides the logic truth table for the RFSW8000.

Table 2. RFSW8000 Logic Truth Table.

SWITCH STATUS		LOGIC CONTROL	
RF1 TO RFC	**RF2 TO RFC**	**VRF1**	**VRF2**
ON	OFF	High	Low
OFF	ON	Low	High

The correct timing of the RFSW8000 logic control and RF input is required to ensure optimal performance and reliable operation. Below is turn on/off procedure.

Transmit Power-On Procedure:

1. Connect Power Supplies in OFF mode (0 V) to VRF1 and VRF2 pin.
2. Apply Positive Voltage 3.0 V on VRF1 (pin 6) and, low Voltage 0 V on VRF2 (pin 4).
3. Apply RF input signal to J1 (RF1 pin 1); measure RF output on J3 (Pin 5 RFC).

Transmit Power-Off Procedure:

1. Remove RF input signal.
2. Set the Power Supply Voltage on VRF1 to 0 V.

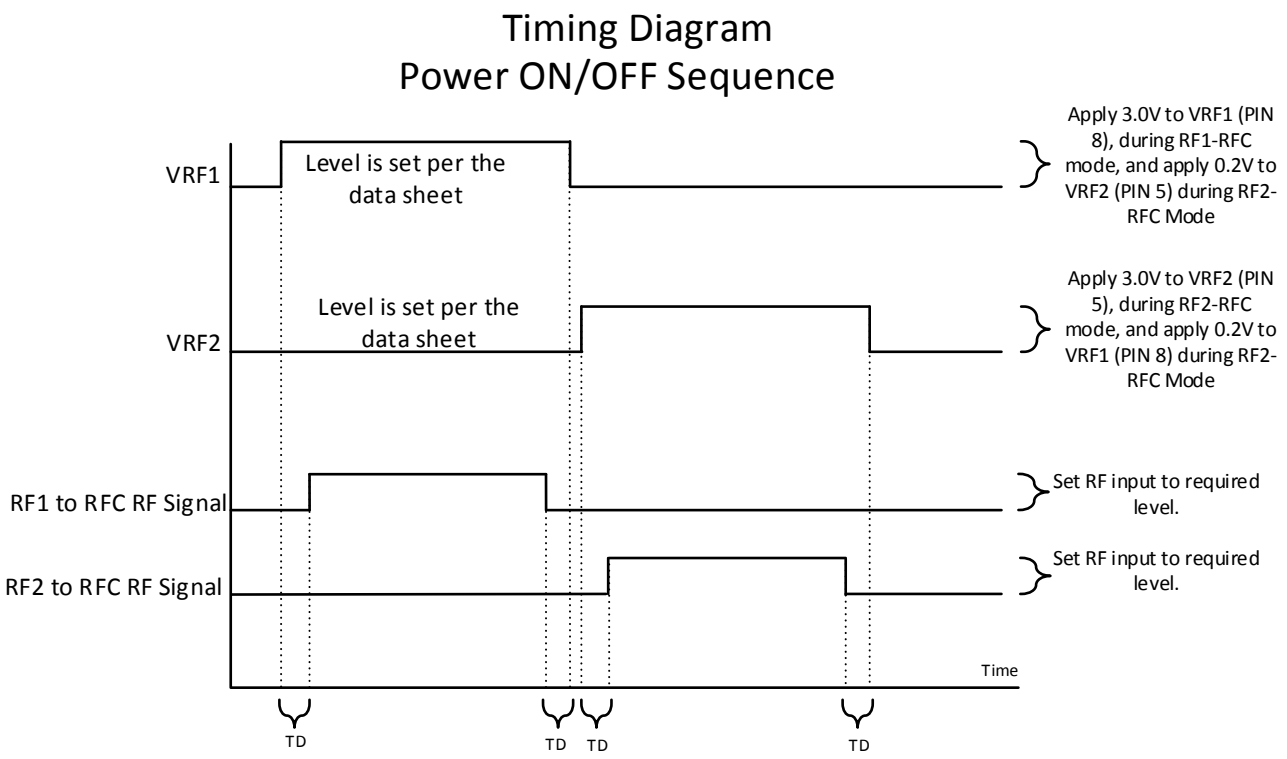


Figure 3. RFSW8000 Timing Diagram

Notes:

1. RF Signal for each specific mode is applied after the DC bias is applied.
2. Total ON/OFF time includes from 10% of control switching to 90% of RF power.
3. TD is defined in the data sheet.

System Architecture Application Circuit Recommendations

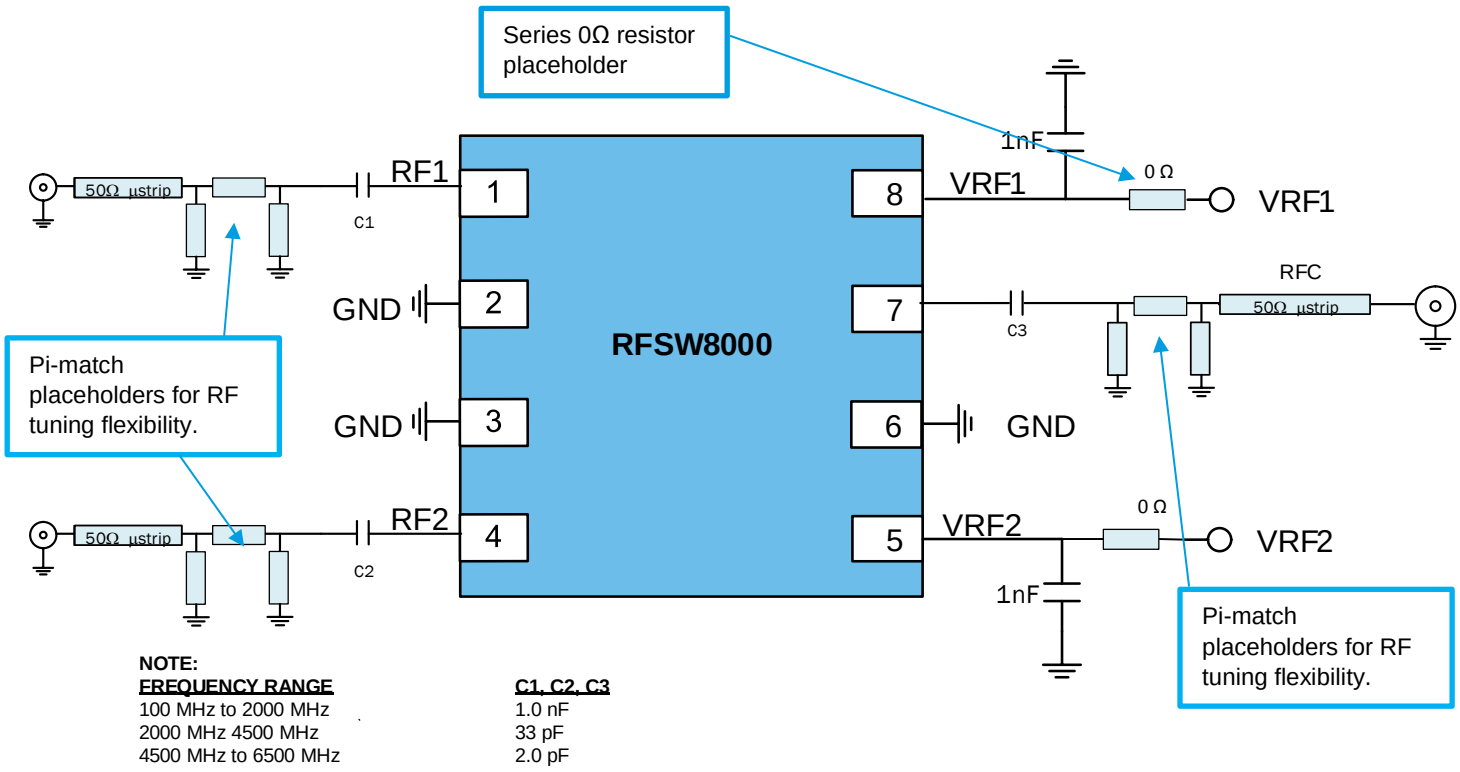


Figure 4. Recommended Application Circuit in a System

1. The above schematic shows recommended bypassing values based on RFSW8000-EVB. Customer should ensure that sufficient bypassing is provided based on their PCB layout. In addition, all bypass capacitors should be placed as close as possible to respective VCONT pins, with the lowest values placed closest to the part pin. It is also recommended that at least one ground via be placed right next to each bypass capacitor ground pad to minimize ground return inductance between the capacitor and the VRF1, VRF2 ground.

2. Requires external DC block on RFSW8000 RF1, RF2, and RFC ports. Decoupling capacitors on VRF1 and VRF2 are optional. Their implementation will depend on system noise present on these lines. For Optimal performance present 50Ω to all RF Ports (RFC, RF1, and RF2 Ports)

3. Recommend keeping a minimum distance of 150um between control lines to minimize coupling. In general, follow Qorvo evaluation board layout as close as possible. Gerber files are available upon request.

4. Route control lines on separate layer, other than the signal layer, whenever possible and isolate controls line traces from (RFC, RF1 and RF2 Ports) and (VRF1 and VRF2) traces. For Optimal performance present 50Ω to all RF Ports (RFC, RF1, and RF2 Ports) decoupling capacitors on VRF1 and VRF2 are optional, but if used should be low noise.

PCB Layout Considerations

Board layout must be carefully considered to achieve optimal performance from any Switch, including the RFSW8000. In addition to providing connectivity between the Switch and external components, the PCB layout is a part of the overall circuit. The RF and DC parasitic of the traces, along with coupling between traces, must be evaluated. The RFSW8000 Evaluation Board PCB layout guidelines provides a good starting point for designing the layout in the actual application.

RF Traces

All PCB traces between the RF pins and matching networks (where applicable) should be 50 Ω controlled impedance lines, as should the traces between the matching networks and the next component in the chain. The RF traces should be routed on top layer to minimize coupling with other RF, control input, and DC traces. If it is not possible for some reasons to route RF traces on top layer, we suggest making sure there is proper isolation between traces on the layout to avoid any coupling issues. RF lines should be isolated from other RF and DC signals by adding solid ground planes (with vias) between them to minimize coupling and cross-talking.
In addition, we also recommend reducing RF trace lengths, wherever possible.

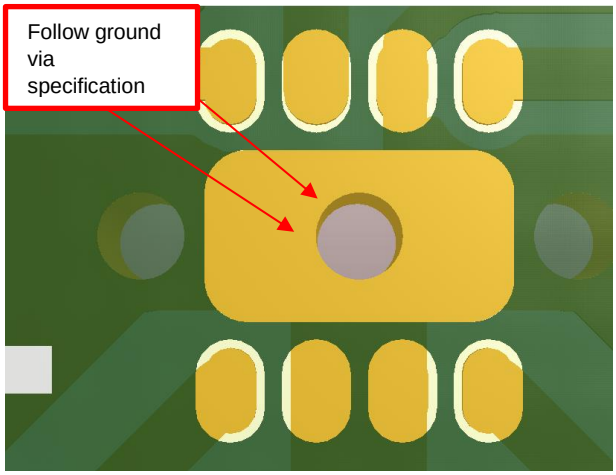


Figure 5a. Recommended PCB layout Considerations

Grounding Considerations

Connect module center ground pad directly to main ground plane layer using as many vias as possible. The PCB ground layer should be close to the component layer, preferably the next layer down to minimize the lengths of via connections between the component and ground layers. Ground paths (under device) should be made as short as possible. This ground layer also provides the reference layer for micro-strip lines.

Close attention should be paid to the grounding of the Switch ground slug, the solid metalized area on the bottom side of the package. This serves as the primary RF and DC ground return for the entire Switch, as well as the primary path for heat removal. A larger number of via holes should be distributed over the entire ground area below the Switch to provide good RF and DC ground returns, as shown in **Figure 5b** below.

Additionally, the vias will serve as a low resistance thermal path through the PCB. Vias passing through multiple copper layers provide the best overall RF, DC, and thermal performance. Ensure proper vias on ground slug / paddle for better thermal consideration. RFSW8000 ground slug / paddle has special electrical and thermal grounding requirements. This pad is the main RF ground and main thermal conduct path for heat dissipation. The GND pad and vias pattern and size used on the Qorvo evaluation board should be replicated. The Qorvo layout files in Gerber format can be provided upon request.

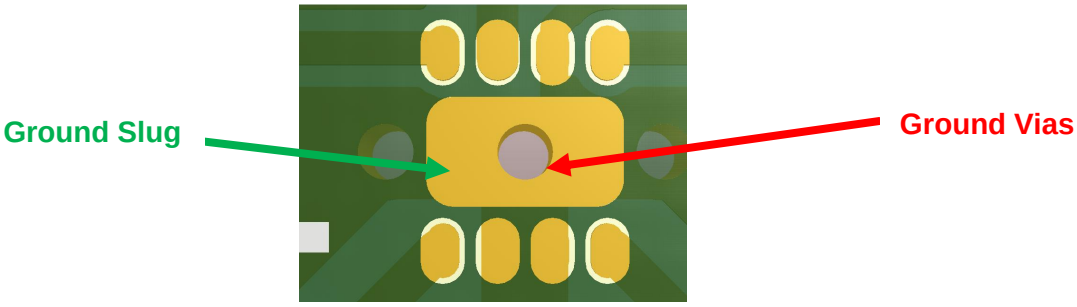


Figure 5b. Recommended Ground Via Placement on Module Ground Slug

PCB Footprint Recommendations

See **Figures 6a and 6b** below for the recommended package outline drawing and solder mask patterns.

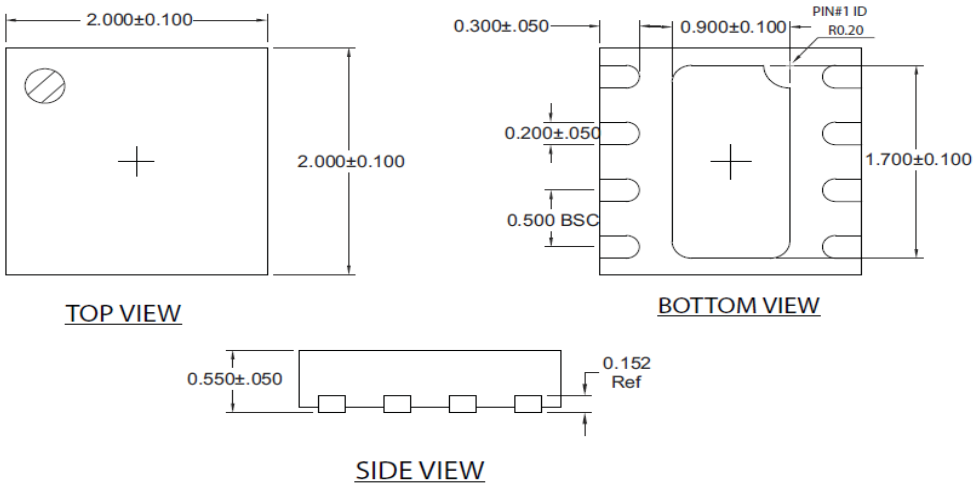


Figure 6a. RFSW8000 Package Outline Drawing

Notes:

1. All dimensions are in millimeters. Angles are in degrees.
2. Dimension and tolerance formats conform to ASME Y14.4M-1994.
3. The terminal #1 identifier and terminal numbering conform to JESD 95-1 SPP-012.

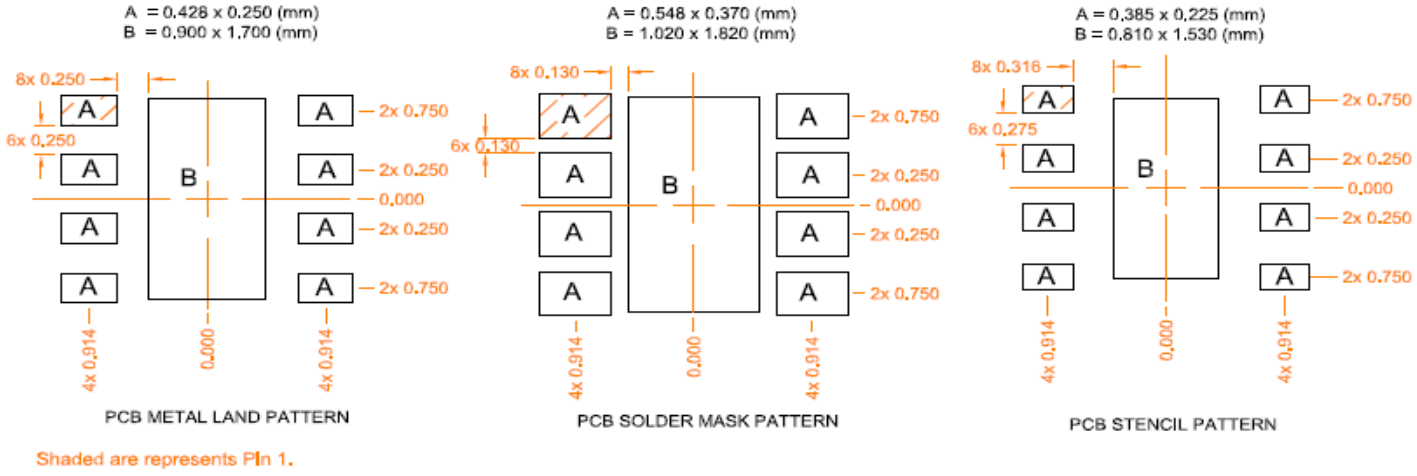


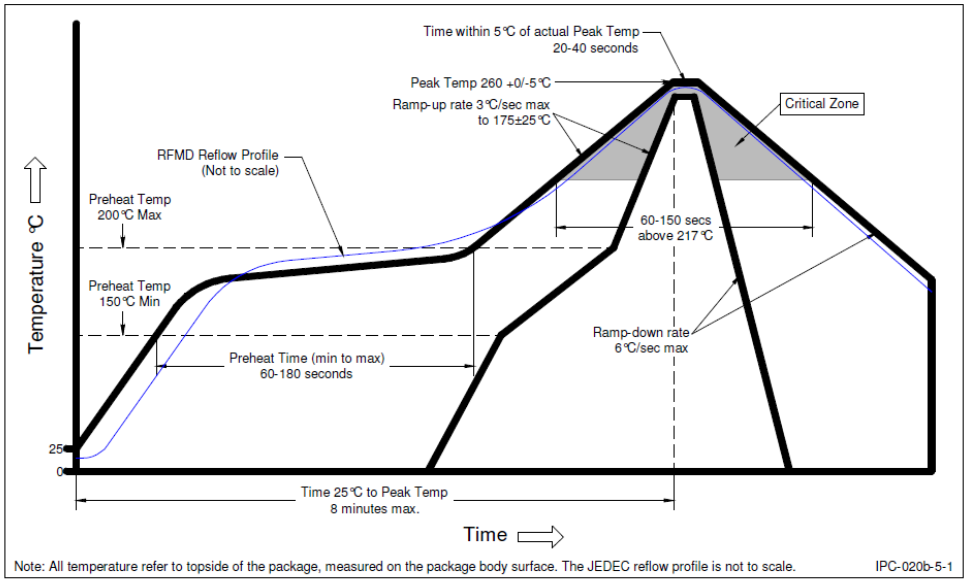
Figure 6b. PCB Footprint Recommended Solder Mask Pattern

Notes:

1. All dimensions are in millimetres. Angles are in degrees.
2. Dimension and tolerance formats conform to ASME Y14.4M-1994
3. The terminal #1 identifier and terminal numbering conform to JESD 95-1 SPP-012
4. Thermal vias for center slug “B” should be incorporated into the PCB design. The number and size of thermal vias will depend on the application, the power dissipation, and the electrical requirements. Example of the number and size of vias can be found on the Qorvo evaluation board layout.

Reflow Profile & Solder Paste

Figure 6c illustrates the recommended reflow profile for the RFSW8000. **Figure 6d** provides solder paste specifications.



CONDITIONS	
Ramp-up rate	3 ºC/second max.
Preheat temperature 175 (±25) ºC	180 seconds max.
Temperature maintained above 217 ºC	60 seconds to 150 seconds
Time within 5 ºC of actual peak temperature	20 seconds to 40 seconds
Peak temperature range	260+0/-5 ºC
Ramp-down rate	6 ºC/second max.
Time 25 ºC to peak temperature	8 minutes max.
Maximum number of reflow cycles	≤ 3
Pre-baking requirements	Refer to JEDEC J-STD-033 if original device package is unsealed
Maximum reflow temperature	260 ºC

Figure 6c. Recommended Reflow Profile & Conditions

Maximum reflow temperature is 260°C. The temperature used to classify the MSL level appears on the MSL label on each shipping bag. Qorvo uses reflow profiles in accordance with IPC/JEDEC J-STD-020 for qualification except for the maximum reflow temperature of 260 °C.

Solder paste used for the Qorvo high temperature reflow qualification.

SPECIFICATIONS	
Solder paste	Multicore 96SCAGS89 (CR39)
Alloy type	Sn95.5/Ag3.8/Cu0.7
Metal content	88.5%
Solder particle size	45 µm to 20 µm

Figure 6d. Solder Paste Specifications.

Support Data

For any further data on RFSW8000, please request Qorvo point of contact such as marketing, sales or representative in your region.

Additional Information

For information on ESD, Soldering Profiles, Packaging Standards, Handling and Assembly, please contact Qorvo for general guidelines.

Contact Information

For the latest specifications, additional product information, worldwide sales and distribution locations:
Web: www.qorvo.com
Tel: 1-844-890-8163
Email: customer.support@qorvo.com

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